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Making Predictions Tangible

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Making Predictions Tangible: Using Data Physicalization to Explore Expectations Around Health Predictions

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Abstract. As predictive technologies like Digital Twin become increasingly applied in healthcare, much focus has been given to improving accuracy, explainability, and trust in health predictions. However, little is known about how people expect to engage with predictions in meaningful ways. In this paper, we explore how people imagine interacting with their health predictions through the physicalization they constructed. We conducted three workshops with 18 design students, using fictional weekly allergy risk predictions and multisensory materials to elicit the embodied, contextual, and creative representations of predicted health risks. Our analysis of the physicalizations revealed that participants envisioned predictions not merely as data points but as embodied and contextual experiences. We identified five sensory mapping strategies and six design themes that illustrate how people engage with predictions. Drawing from these findings, we reflect on design implications to support agency, adaptability, and personally meaningful interactions. We contribute an empirical understanding of users' interpretations and engagements with health predictions, offering inspiration for expanding the interaction design space of future predictive health technologies.

Keywords: Health Predictions · Data Physicalization · Workshop · Interaction Design · Digital Twin.

1 Introduction and Background

Predictive technologies are widely embedded in our everyday life, from weather forecasts, traffic congestion prediction, to potential health risks. In healthcare, technologies like risk calculators [1] and Digital Twins [3, 7] are increasingly used to forecast health conditions, such as blood glucose trends [4] and allergy flare-ups [7]. These technologies not only support early disease detection and decision-making for clinicians but also help individuals anticipate health risks and take proactive actions.

However, in practice, users often struggle to make sense of predictive results, not only due to the complexity of numerical scores, charts, and graphs, but because these results may lack explanation, transparency, and relevance to people's

daily lives [16, 17]. To address these, many studies have focused on enhancing the accuracy of predictions (e.g., integrate more factors into predictive models [17]), clarifying how predictions are generated (e.g., explainable AI [16, 15]), and designing better ways to communicate uncertainty (e.g., visualizations like icon arrays or dotplots [6, 5, 9]). Yet, accuracy and explainability alone do not guarantee users' trust in predictions. People may lose trust after a single mistake in the prediction [12] and require a sense of agency over the predictions [14]. Besides, previous studies in domains like diabetes and mental health have shown that users react differently depending on how predictions are presented (e.g., visual, textual or hybrid [15]) and framed, which can affect their motivation, behavior and anxiety [4, 11, 12]. These findings point to the need to understand how people expect to interact with predictions.

In response, we explored people's imaginations and expectations to interact with predictive health information through constructive data physicalization workshops. Constructive data physicalization, where participants represent data through physical artifacts [13], offers an opportunity to elicit people's mental models and values in their creation. Previous research has shown that encoding data in a physical form can foster reflection, sense-making, and personal connection to abstract information [13, 10, 2]. In our workshops, participants were encouraged to physicalize the fictional respiratory allergy risk predicted from a personal Digital Twin [7], by using a variety of sensory materials (visual, sound, touch, smell, taste). A personal Digital Twin in this context refers to a data-driven model that constantly integrates physiological and environmental inputs to make predictions for individual's health [3]. Rather than evaluating how accurately the predictions are mapped into physical forms, we used it as a lens to elicit interaction expectations and embodied or material metaphors beyond numerical data points.

This paper aims to provide design insights into technologies that communicate health predictions by offering two contributions. First, it presents empirical insights into how people use sensory modalities and metaphors to materialize predictive health information, revealing five sensory mapping strategies and six design themes. Second, it outlines the design implications derived from these themes for future interaction design with health predictions.

2 Method

2.1 Participants

We recruited 18 participants between 21 and 32 years old (M=26.67, SD=3.01; eleven self-identified as men, six as women, and one as self-described). All participants were students with a background in industrial design, recruited through the university network. They had prior experience participating in co-design activities. None had food-related allergies or restrictions related to any materials given in the workshop. The study was approved by the University Ethical Review Board, and informed consent forms were obtained before the workshop.

2.2 Procedure

The 90-min workshops were conducted in an indoor space with three tables and a display. Participants completed a consent form and a brief demographic questionnaire. The workshop followed four phases:(1) Introduction (10 min): We introduced the concept of constructive data physicalization — representing data with tangible materials — and framed the study around fictional respiratory allergy risk predictions generated by a personal Digital Twin [7]. (2) Material Engagement (5 min): Participants explored a variety of sensory materials (e.g., essential oils, foam balls, yarns, marshmallows) to spark ideas for multisensory representation. (3) Two Activities (2×20 min): Each pair constructed physicalizations based on the scenarios and datasets. The second iteration used updated predictions, while participants could either build on the previous one or redesign a new one. (4) Self-Interview and Discussion (2×10 min + 10 min): After each activity, each pair completed a self-interview and at the end a group discussion was held to share and reflect on the designs. Participants audio-recorded their making process and video-recorded their self-interviews.

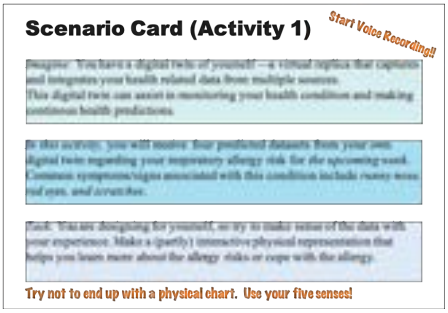
2.3 Materials

Crafting materials. Building on material selections from previous physicalization workshops [8, 10], we provided a range of materials that afford multisensory engagement (Table 1) to support the creative process. The open space around the tables was also included as the work area. Participants were explicitly encouraged to use five senses as a way to surface possible metaphors and embodied associations around prediction, expanding how predictions could be experienced beyond conventional visual or verbal forms. An image of the crafting materials is included in the Appendix Fig. 2.

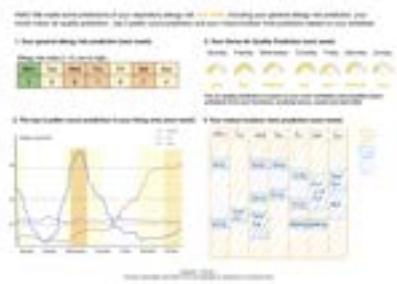
Table 1: List of Materials Categorized by the Main Sense they Support

Category	Materials
Sight	Multicolor pencils, multicolor A4 papers, multicolor crepe papers, post-its
Touch	Playdoh, multicolor yarns, multicolor twist sticks, baking paper, toilet paper, felt balls, foam balls (5 sizes)
Sound	Aluminum foil, plastic bags, rubber bands, plastic cups (2 sizes), paper cups, chewing gum containers (3 types: paper box, plastic container, metal container)
Taste	Lemon juice, water, wasabi, ketchup, candy (2 types: Haribo sour spaghetti, Haribo Duo's fruity), marshmallows, chewing gum (3 types: strawberry, eucalyptus, mint)
Smell	Cooling balm, Sichuan pepper, cardamom powder, dried coriander, chili powder, coffee powder, tea (5 types: rooibos, fruit, lemon, green, chamomile), essential oils (6 types: lavender, pepper mint, lemongrass, eucalyptus, sweet orange, tea tree)
Building Tools	Pins, paper clips, rulers, tapes, glue, scissors, tea bags, cotton pads, paper plates, wooden blocks, wooden spoons, wooden sticks, coffee stirrer, hand sanitizer

Scenario Cards. Each activity was supported by a scenario card inspired by Huron et al. [8]. In activity 1, participants were required to interpret a 1-week respiratory allergy prediction from a Digital Twin and build a (partly) interactive physicalization. In activity 2, they were given an additional week of predictions and expected to build on their first design or create a new artifact to convey these continuous predictions. The full sets of scenario cards can be found in the Appendix Fig. 3.



(a) Scenario in Activity 1



(b) Datasets in Activity 1

Fig. 1: Example scenario card and datasets, see Appendix Fig. 3 and Fig. 4 for the full contents

Printed Fictional Datasets. For each activity, we provided an A4 sheet including four visualizations of allergy predictions: (a) daily risk-index table, (b) home air-quality table, (c) top-3 pollen count line chart, and (d) indoor/outdoor schedule calendar. These formats draw on common pollen-risk and home-IoT visualizations research on Digital Twin allergy [7], as well as Lallemand and Oomen's approach [10]. In Activity 2, all prediction values were updated to indicate the next week's predictions. See full datasets in the Appendix Fig. 4.

Printed Self-interview Card. After each activity, each pair completed a 10-min video-recorded interview, immediately guided by a self-interview card. They explained their physicalization and reflected on their process, without being affected by other groups' creations and interpretations. As the results are not analyzed deeply at this stage, the interview prompts are included in the Appendix Fig. 5 for reference.

3 Results

18 physicalizations were produced by 9 paired groups during the workshops. At this stage, we analyzed the sensory modalities and materials participants used in their physicalizations, along with the verbal explanations from self-interview. We mapped their sensory usage to the physicalization and extracted design themes from their explanations through thematic analysis. The pictures, mapping and explanations for each physicalization are presented in the Appendix table 2. We refer to each group as GX(e.g., G2, G3).

3.1 Sensory and Material Mappings to Predictions

Most of the groups produced a single physicalization incorporating multiple sensory interactions. Three groups created several physicalizations to communicate different predictions. We analyze which senses and materials were used and why, with details in the Appendix table 2.

Sight as the dominant usage. Every group used visual encoding, including color, shape, size, amount, length, distance, angles, blank space, or spatial layout, to communicate predictions. Moreover, the brightness of the color in G2's example indicates the selected day, while an inconsistent object was used to bring attention to the abnormal activity, which shows people's need to notice critical information at a glance.

Smell as embodied sensation. Almost every physicalization used some olfactory interaction, not as mere decoration but as an active data channel for "sniffing" future risks. Lemongrass, lavender as "safe" scents (G5), coffee as a "moderate" level of risk (G6), eucalyptus (G9), garlic and chili powder (G6) as "warning" scents. By choosing scents that felt intuitively "good" or "bad", the participants showed that they expected the predictions to be personal and resonate with their own feelings. Several groups mapped the predicted risk level to the scent intensity (e.g., G5's foam balls diffused a stronger lemongrass scent for higher indoor air quality risks), while others connected distinct smells to different predictions (e.g., G3 used lavender for a general allergy index and eucalyptus on marshmallows for indoor risks). Some groups mixed food powders on cotton pads to indicate specific pollen types, such as dried coriander for "grass" pollen, tea tree oil for "tree" pollen, and lemon grass oil for "ragweed" pollen (G2). Beyond direct encoding, scent was also layered atop other modalities (e.g., G1 added essential oils to playdough to emphasize the predictions), or used peripherally to attract people's attention to details (e.g., G7 added spices on top of foam and felt balls).

Touch for temporal engagement. Participants used tactile interactions to "grasp" predicted risks, including rubbing, lifting, and sliding elements to bring tomorrows' predictions to the present. G2 created two water bags for users to feel predicted pollen particles through tactile feedback while squeezing, while G3's pull-string mechanism allowed users to lift the cord up and down to scroll over hourly predictions. G4 and G5 leveraged contrasting textures (firm foam balls versus soft felt balls) to distinguish different pollen types, so that touching distinct surfaces indicated respective allergens. Meanwhile, G9 introduced a textured string whose spikes grew sharper for higher risks and denser for predictions further in the future, allowing users to sense both the severity and the temporal distance of predictions at once. Similarly, G7 used the tension of the string to mark predicted risks, but it is described as a temporal interaction because users can easily read the tension visually.

Taste as surprise and hidden insight. Mapping predictions to flavors makes it a private ritual that uncovers "hidden" truths about users' upcoming week. G1 invited people to understand "the hidden risks" in the coming days with the "surprise taste of candy" from the marshmallow barbecue sticks. To ensure people can still revisit the predictions, a package containing several marshmallow barbecue sticks with the same flavor distributions was planned to be given weekly. Similarly, G9 integrated prediction into the morning ritual of coffee: a spicy blend of chili and Sichuan pepper marked days of high predicted risks, cardamom-infused beans signified moderate, and a naturally sweet, fruity

roast indicated low. By choosing flavors that resonated emotionally, participants made the process of revealing predictions playful and personal.

Sound for ambient cues. By translating predictions into everyday noises, participants imagined predictions as ambient cues of recommendations. G6 used crinkling paper to mimic rustling leaves ("not go outdoors"), while the sound of air through marshmallow-filled tea bags resembled an air conditioner's hum ("indoor is safe"). A gentle jingle of lemon-juice-topped cups evoked familiar domestic sounds such as dish washing, strengthening the "indoor" recommendation. G9 wanted to communicate the predicted pollen types with a "symphony" representing three pollens, which were represented through the cups containing crepe papers, wooden blocks and aluminum foil balls respectively.

3.2 Design Themes: How People Physicalize Predictions

As our second activity asked participants to physicalize not only next week's predictions but also the week after, we examined how they extended their initial 7-day timelines (activity 1) into a continuous stream of predictions. In activity 1, almost every group created a simple weekly view, either by labeling each day (M, T, W...) or by repeating a set of elements seven times. Only two groups (G2, G7) created a default daily view of the predictions. In activity 2, we found six design themes emerged to physicalize predictions.

Theme 1: Scale up predictions with consistency. G4 regarded it as an additional weekly prediction, explaining that another seven plates of items would represent daily predictions for the week after next. G8 also planned to show a seven-day prediction but emphasized the importance of consistency and reusable elements in the design, so they created a seven-day view of the same set of environments, each with different amount of coffee powder around as predicted risks. G5 and G6 chose to use other modalities for data in the different week.

Theme 2: Expand predictions through spatial variations. G5 added spatial qualities to their initial weekly prediction chain and turned it into a shape-changing object for people to display at home. G6 experimented with various materials to make sounds for two types of prediction in the week after next week as a comparison of using color and smell for predictions in their first construction.

Theme 3: Enable destruction and reflection of predictions. G1 designed the edible marshmallow strings to be delivered as a package with several same ones inside every week, aiming to allow reflection, destruction of data, and observing whether the data have been consumed or noticed by others. The actual taste was also intentionally hidden behind the plain marshmallows, which was described as a "secret".

Theme 4: Shift from numbers to impacts. With more predictions made for the distant future, G2 chose to show its impact on future activities using a spinning wheel of time and red cups as indicators for some affected activities. While, G3 added a dynamic mechanism of time to their design, allowing people to see the changing impacts throughout the day for different days, which we interpret as their way of capturing the dynamic nature of predictions.

Theme 5: Support shared predictions among family. The continuous prediction of pollen counts in the environment reminded G7 to care about the access to family members' health predictions and how the environmental predictions should be noticed by every member at home. Therefore, they added smell as a peripheral way to interact with and allow access to different family members' health risks, turning it into a household ritual.

Theme 6: Present predictions with uncertainty. Lastly, G9 considered the uncertainty of further predictions, so they crafted a confidence interval of the predictions and an abstract tactile representation of future predictions. The further the prediction, the denser the representation became.

4 Discussion

4.1 Design Implications

Our study offers possible design directions for future predictive health technologies by exploring how interactions with predictions can be reimagined through embodied, contextual, and multisensory representations. While the findings are shaped by our guided and speculative workshop setting, which encouraged participants to move beyond visual and numerical forms, they reveal underexplored interaction metaphors and sensory strategies for design.

In our workshops, participants used sensory and material strategies to represent ambiguity of prediction. Tactile fuzziness conveyed uncertainty, fading density suggested future distance, and olfactory intensity expressed urgency for awareness. Although these mappings were encouraged by the multisensory materials we provided, they illustrated how embodied representations can represent predictive ambiguity in subtle and intuitive ways. It resonates with previous work which used quantile dotplots to facilitate the understanding of uncertainty [9] and extends it to ambient and haptic modalities and metaphors. Less commonly used modalities, such as smell and taste, also emerged, such as lavender for safety, chili for danger, or a surprise coffee sip marked a personal reveal moment. While we do not claim these as preferred channels, their speculative use suggests how people relate to predictions through embodied metaphors, which were also seen in previous sensory interactions [2, 10]. These metaphors suggest that predictive technologies could also support affective interpretations and personal sense-making, beyond traditional numerical forms, aligning with previous work on explainable AI which calls for representations tailored to users' cognitive and affective needs [15, 16].

Many physicalizations also reflected the expectations of agency in how predictions are accessed and shared. For example, edible predictions point to the need for people's agency to protect predicted personal insights, allowing explorations of hidden insights, and interaction rituals for daily "reveals", which could be used in design to promote engagements. One participant used ambient scent to share predictions within family without exposing sensitive details. It raises the question on how predictions should balance the sharing with privacy, retaining users' control over details.

Finally, participants experimented with different time presentations of predictions, such as sliding mechanisms, spatial layouts, or abstract representations of far future. These designs suggest supporting flexible zooming across time and shifting between detail to overview in predictions, thereby expanding the possibilities beyond fixed linear timeline-based health interfaces.

Overall, rather than proposing specific design solutions, we treat these physicalizations as design probes that reflect underexplored interaction possibilities, such as ambiguity, ritual, agency, and flexible timelines, for the future design of predictive health technologies.

4.2 Limitations

This study has several limitations. First, we explored health predictions only through the lens of predicted allergy risks, which may not represent other types of prediction, such as those related to behavior change or lifestyle. As a result, the metaphors and embodied values in the physicalization are inevitably affected by how people perceive "risk". Secondly, while fictional datasets and scenarios in a controlled setting ensures the comparability among users, participants might generate richer, more personally resonant designs if they had worked with their own health data or in a longitudinal field study. Third, although we provided a wide range of materials covering multiple senses to spark imagination, these materials were not intended to be a validated design toolkit. Additionally, our participants were design students with co-design experience, which distinguishes them from the general population. Their familiarity with creative design approaches likely encouraged more expressive and speculative outputs, which may not reflect how lay users would approach predictive health information and offer generalizable insights. Finally, this study did not address the potential pitfalls of predictive technologies, so our findings do not capture how people might react to incorrect, conflicting, or failed predictions. Prior work suggests that even a single inaccurate prediction can undermine user trust or trigger algorithm aversion [12], which requires further explorations on how people expect predictive technology or system to acknowledge and communicate such aspects.

5 Conclusion and Future Work

This study represents an initial step toward understanding how to design interaction with health predictions generated by Digital Twins, with implications that may extend to other predictive health technologies. Through physicalization workshops, we explored how participants make sense of and materialize health predictions. Our reflections on sensory mappings and design themes reveal possible implications to support agency, adaptability, and meaningful engagement with predictions, while treating multisensory creations as a lens to understand user expectations. Future work could build on these insights by involving broader participant groups (e.g., end users and experienced designers) and engaging with

real-world health data to deepen understanding of how predictions are interpreted and expected in everyday life. As predictive technologies become more pervasive, questions of privacy, ethics, and interpersonal boundaries will require further design attention, especially in domestic or shared environments.

6 Acknowledgment

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Appendix

Fig. 2: An overview of the crafting materials used in the workshops

Table 2: Data Physicalization Outcomes and Data Mapping

Group	**Physicalization** Top: activity 1 Bottom: activity 2	**Data Mapping**	**Senses for Interaction**
G1		Playdoh with different colors and heights represent the general allergic risks index during the week. The same information is emphasized by the smell of different spices and essential oils on top of the playdoh. Time is written as the capital letter on the paper. The spatial line is created with twist stick and some threads to indicate the trend of the general allergic risk.	Sight, Smell

Table 2 Continued from previous page

Group	**Physicalization** Top: activity 1 Bottom: activity 2	**Data Mapping**	**Senses for Interaction**
		Candy bundles or pocket candies matched colors with their first physicalization. To ensure that people can still revisit their data after tasting it, they envision people will get a box or a bundle of same candies for every day. The wooden marshmallow stick is their final physicalization. The flavored candies were inserted into the marshmallow to indicate risk levels on days of the week. The way in which the candies were hidden from the surface is how they embody the hidden allergic risks. People are considered to receive several strings of wooden sticks with same flavor distributions every week.	Sight, (Touch), (Smell), Taste
G2		The color of the water bags differentiates indoor (yellow) and outdoor (green) information. The particle amount in the water bag indicates air quality indoors and pollen amount outdoors. The size and number of balls represent three top pollens, which can be observed visually or by touch. The hour of the day is marked by a number on the ruler. The brightness of the water bags suggests whether to go outside or stay home. Three different smells link to the top pollen risks: dried coriander for grass pollen, tea tree oil for tree pollen, and lemongrass for ragweed pollen.	Sight, Touch, Smell
		Red cups highlight schedule changes, with red indicating minor changes and paper cups indicating major changes. Schedules and activities are shown on a small purple triangle with pink papers representing different days. Rotating it changes the daily view, like a casino wheel, with more resistance indicating approaching a highlighted time. Color, smell, brightness, and tactile cues remain consistent.	Sight, Touch, Smell

Table 2 Continued from previous page

Group	**Physicalization** Top: activity 1 Bottom: activity 2	**Data Mapping**	**Senses for Interaction**
G3		Each double-marshmallow string indicates a day in a week. Only Monday and Tuesday are represented to show the idea here. The space in-between marshmallows are the daily timeline. The amount and color of string around the marshmallow represents different types of pollen counts daily. Lavender smell is chosen to indicate the general allergy risk index while eucalyptus as indication for indoor air quality. The smell intensity of the lavender from the marshmallow links to the general allergy risk index, the stronger the higher risks. A stronger eucalyptus smell indicates a higher indoor air quality.	Sight, Smell
		They built on the previous design by giving some adjustability of the string to dynamically represent time. Each string can have its length change dynamically according to the time of the day and leads to the height change of two marshmallows, which indicates the air quality predictions at different times of a day. If both are high, it means very good air quality. Both are down means bad. The daily general allergy risk is indicated by the intensity of the lavender smell of the string. The amount of tea leaves inside the bag attached to the marshmallow indicates the indoor air quality (the higher the number, the worse). The color and texture of twist sticks indicates pollen type, and rotations indicate pollen amounts, linking to everyday pollen allergy risk outdoor.	Sight, (Touch), Smell

Table 2 Continued from previous page

Group	**Physicalization** Top: activity 1 Bottom: activity 2	**Data Mapping**	**Senses for Interaction**
G4		It shows Monday and Tuesday by the twist sticks. The capital letter's color and the type and intensity of the smell mean the general allergy risk of the day. The things inside the plates are outdoor predictions, including the type of balls indicating the type of pollen and the amount as the pollen count. The barbecue string indicates suggestions for activities on that day. The unclear prediction for the working place is shown with a single-colored playdoh, while the occupied area on the cube indicates the allergy risk of doing certain activities. Things inside the cup indicate indoor predictions; the more, the worse.	Sight, Touch, Smell
		More plates will show up to match the continuous data input. They created a detailed version of the previous one for Monday, Tuesday, Wednesday, and Friday. The amount and color of crepe paper straps are added as a more obvious general indication of allergy risk.	Sight, Touch, Smell
G5		The white balls are given different essential oil smells to indicate indoor air quality (e.g., lemon grass for 7, lavender for 2). The balls together compose the weekly prediction chain with angles between the balls that indicate the general prediction of allergy risk (e.g., 30 degrees for 3). In the red cup there is a representation of three types of pollen using ketchup, chili pieces and coriander to create different textures of the foam balls.	Sight, (Touch), Smell

Table 2 Continued from previous page

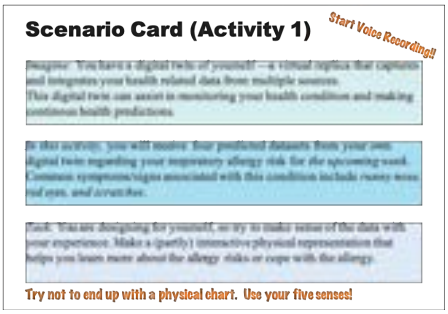
Group	**Physicalization** Top: activity 1 Bottom: activity 2	**Data Mapping**	**Senses for Interaction**
		The spatial dimension is added to the chain, making it a 3D shape-changing structure. The first weekly prediction is shown on the front structure, and balls after indicate the prediction of the week after next week. The color of the threads between the balls of the next week and the following week indicates changes in home air quality. Blue means it will be better than the previous week and orange means worse. If there is no change, the same ball is used to connect the others.	Sight, (Touch), Smell
G6		The color and size of the playdoh on top refers to the general allergy risk. The different smells in marshmallows also indicate the general risk of allergies (lavender for good, garlic for bad, and coffee for moderate). The little balls on the playdoh show the amount of pollen. Indoor and outdoor time are represented using the colors of the twist sticks on top. The number and lengths of the twist sticks show the length of time spent outdoors or indoors. The time at home is represented by fluffy white twist sticks. In the office it is spiky dark blue and in the outdoors it is light blue.	Sight, Touch, Smell
		They added how to make choices to stay home or go outside for the week after the week using sound. The louder the sound, the less is suggested. Marshmallows inside tea bags indicate home air quality, mimicking the sound of a ventilation system when blown. Listening to the sound of lemon juice made while shaking cups is iterated from blowing marshmallows in the bags as an easier way to represent home air quality. The louder the sound, the less it suggested we should stay home. Using paper inside the plastic bag to create sounds like leaves outside to indicate the outdoor pollen allergy risks. The louder the sound, the less we were suggested to go outside.	Sight, Sound, Smell

Table 2 Continued from previous page

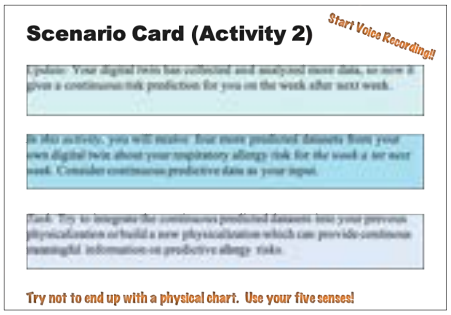
Group	**Physicalization** Top: activity 1 Bottom: activity 2	**Data Mapping**	**Senses for Interaction**
G7		The color of the paper indicates outdoor (pink) and indoor (blue). The lemon juice bottle together with the blue yarn ball works as a personal avatar. The daily view can change by spinning the black essential oil bottle. The flow of the red line indicates the daily schedule of going outside and moving inside. The tension of the red line can be changed by the shape of the surface to indicate the risk level during different scheduled activities. The white objects with three different sizes on top of the pink paper indicate the types of pollen. The distance to the line indicates the relevance or impact of the outdoor activity, and its tension can react to the outdoor activity that people touched on the red line.	Sight
		More colorful yarn balls were added to indicate family member's prediction at home. The group linked the smell of three balls to the predictions of the top 3 pollen counts, e.g., coriander smell for grass pollen, tea tree oil smell for tree pollen, and rooibos tea smell for ragweed pollen. The mixed smell is considered an invitation for people to come closer and take a closer look at the data.	Sight, Smell
G8		Different playdoh shapes indicate risky activity or environment people may encounter on that day of a week, e.g., sofa->indoor, computer->office, water with sand/umbrella->beach, tree->forest. The amount of coffee powder indicates pollen or indoor pollution. If the scheduled activities on that day do not pose allergy risks, it is indicated with water and a green tick (top left).	Sight, Smell

Table 2 Continued from previous page

Group	**Physicalization** Top: activity 1 Bottom: activity 2	**Data Mapping**	**Senses for Interaction**
		The group tried to make some consistent, adaptive and reusable creations for every day. In this new set of physicalization, every day, there is a representation of each environment with or without coffee powder around. The strongness of the smell and the amount of coffee powder indicate the level of risk in that environment.	Sight, Smell
G9		The color of the candy bridge represents the home air quality prediction every day in the week. Every day, the taste of the coffee attached to the bridge changes according to the general prediction of allergy risk. Low risk is fruity sweet coffee, moderate with cardamom spice, high is coffee with cumin, chili and pepper. The colorful playdoh string is a representation of daily activity and how risky people may experience. When people touch the playdoh string, mixed sounds from three red cups can be produced to indicate three types of pollen. The little cup with colorful threads uses the smell in different parts of the thread to indicate the risk level during the day (lavender low risk, eucalyptus quite risky).	Sight, Sound; Sight, Smell; Taste
		The group added one more textured string for coming risk predictions. The spikier, the higher the risk, and the further the prediction, the more density people can feel. Another orange playdoh with changing width was created to indicate the confidence interval of the predictions.	Sight, Sound; Sight, Smell; Taste; Sight, Touch;

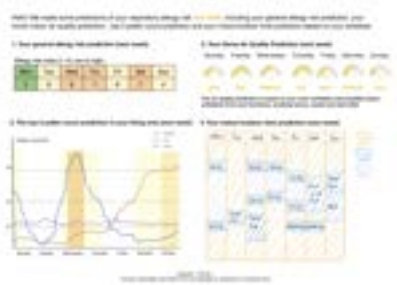


(a) Scenario in Activity 1

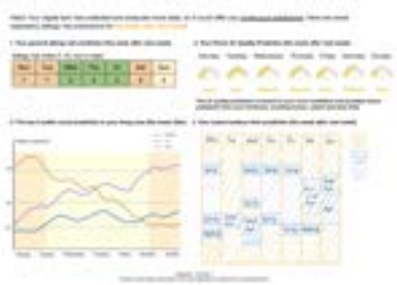


(b) Scenario in Activity 2

Fig. 3: Scenario Cards



(a) Datasets in Activity 1



(b) Datasets in Activity 2

Fig. 4: Fictional Datasets

(a) Self-interview Card in Activity 1

(b) Self-interview Card in Activity 2

Fig. 5: Self Interview Cards

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